which is the diametric opposite of what we wanted. Unlike the **X** operator, **D** is not Hermitian. Instead, it satisfies

$$\mathbf{D}^\dagger = -\mathbf{D}.$$

An operator with this property is called *anti-Hermitian*.

Although anti-Hermitian and Hermitian operators are opposites, it's very easy to go from one to the other. All you have to do is multiply by the imaginary number i or $-i$. Therefore we can use **D** to construct an operator that *is* Hermitian, namely

$$-i\hbar\mathbf{D}.$$

If we look at the action of this new Hermitian operator on wave functions, we find that

$$-i\hbar\mathbf{D}\psi(x) = -i\hbar\frac{d\psi(x)}{dx}. \qquad (8.10)$$

Keep this formula in mind. It will soon play a leading role in defining a very important property of particles—their momentum.

8.2 THE STATE OF A PARTICLE

In classical mechanics, the "state of a system" means everything you need to know to predict the system's future, given the forces acting on it. That, of course, means the positions of all the particles comprising the system, as well as the momenta of those particles. From a classical perspective, the instantaneous positions and momenta are entirely independent variables. For example, for a particle of mass m moving along a one-dimensional axis x, the momentary state of the system is described by the pair (x, p). The coordinate x is the location of the particle, and $p = m\dot{x}$ is its momentum. Taken together, these two variables define the phase space of the system. If we also know the force on the particle as a function of its position,